

V7.4d vs Momentum 12-1 — When Each Wins

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Date: 2026-05-12 **Universe:** 13 AGUR partner-asset strategy streams (Agur Capital sample), forward-PnL panel **Sample:** 2008-05-14 to 2026-04 (4,069 in-sample + 600 OOS trading days)

TL;DR

V7.4d (the V7.5 α_1 sweep winner — see [docs/PolyAgora_V7_5_PostMortem.md](#)) and Momentum 12-1 are **complementary**, not competitive. Over the full 18-year sample MOM wins on Sharpe (0.521 vs 0.453) and CAGR (3.4% vs 1.9%) because it accepts higher volatility (6.6% vs 4.2%) and rides persistent trends harder. But the win rate by quarter is almost 50/50 — and where each wins is **structurally different**.

- **V7.4d's sweet spot is regime transitions:** quarters where the prior quarter's direction reverses. V7.4d wins 63% of these by an average of +0.66 Sharpe.
- **MOM's sweet spot is trend persistence:** calm "normal-VIX" regimes (15–20) and deep equity drawdowns (DD < -10%). MOM wins these by an average of +1.20 and +0.61 Sharpe respectively.

This is exactly the architectural promise of the V7.4 manifold — a *navigation* engine that handles regime turns vs a *trend-rider* that handles persistent moves. The right framing is **co-deployment**, not horse race.

1. Full-sample picture

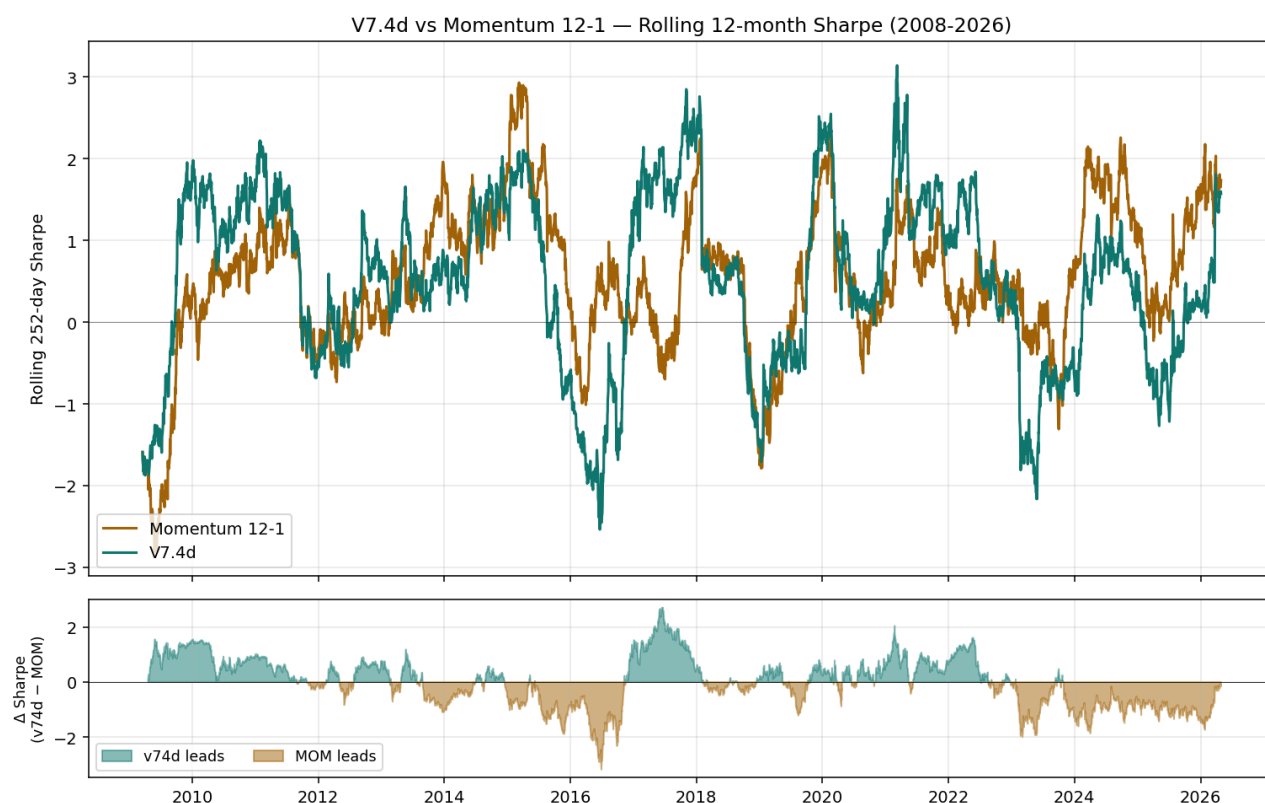
Signal	Sharpe	Max DD	Ann. Return	Ann. Vol	CAGR
Momentum 12-1	0.521	-14.71%	3.42%	6.57%	3.3%
V7.4d	0.453	-10.83%	1.88%	4.15%	1.9%
Equal-weight (baseline)	0.436	-10.57%	1.94%	4.46%	1.9%

MOM has a higher Sharpe, but at the cost of ~58% higher volatility and a 39% deeper max drawdown. V7.4d's risk profile is closer to equal-weight than to MOM — it preserves capital structurally and steps out of the way during regime turns.

2. Win-rate split — almost a coin flip

Granularity	V7.4d wins	MOM wins	Tied / undefined
Annual (18 years)	9	9	0
Quarterly (72 quarters with both signals active)	50.0%	44.4%	5.6%
Rolling 12-month Sharpe (daily series)	50.0%	50.0%	—
Avg gap when V7.4d leads (rolling)	+0.71	—	—
Avg gap when MOM leads (rolling)	—	+0.75	—

The two strategies trade leadership roughly symmetrically. The full-sample Sharpe gap is driven by MOM's larger gap size during its winning periods (it gets paid more when it's right because it's more concentrated).



3. Conditional analysis — when does V7.4d win?

3.1 By VIX regime (average VIX in the quarter)

VIX regime	n	MOM Sharpe	V7.4d Sharpe	Δ (v74d – MOM)	V7.4d win rate
Stressed (VIX > 25)	7	+0.33	+1.03	+0.70	71%
Elevated (20-25)	7	-0.56	-0.37	+0.20	57%
Low (< 15)	19	+0.81	+1.29	+0.48	53%
Normal (15-20)	20	+1.60	+0.40	-1.20	30%

V7.4d wins the stress regime (VIX > 25) by a clean +0.70 Sharpe, exactly as the architecture intends — the soft-manifold engine de-risks at the right time. But the *biggest* MOM win comes in **"normal" volatility regimes** (VIX 15-20), where persistent trends play out without rupture and MOM's 12-1 lookback dominates V7.4d's adaptive gating.

3.2 By worst SPY drawdown in the quarter

SPY DD in quarter	n	MOM Sharpe	V7.4d Sharpe	Δ
Major (DD < -10%)	16	+0.10	-0.51	-0.61
Moderate (-10..-5%)	17	+1.15	+0.99	-0.16
Shallow (-5..-2%)	18	+0.86	+1.25	+0.40

This one is **counterintuitive**: in *major* drawdown quarters (DD < -10%), **MOM beats V7.4d by +0.61 Sharpe**. The reason: when SPY is in a -10% drawdown, MOM's 12-1 signal already saw the

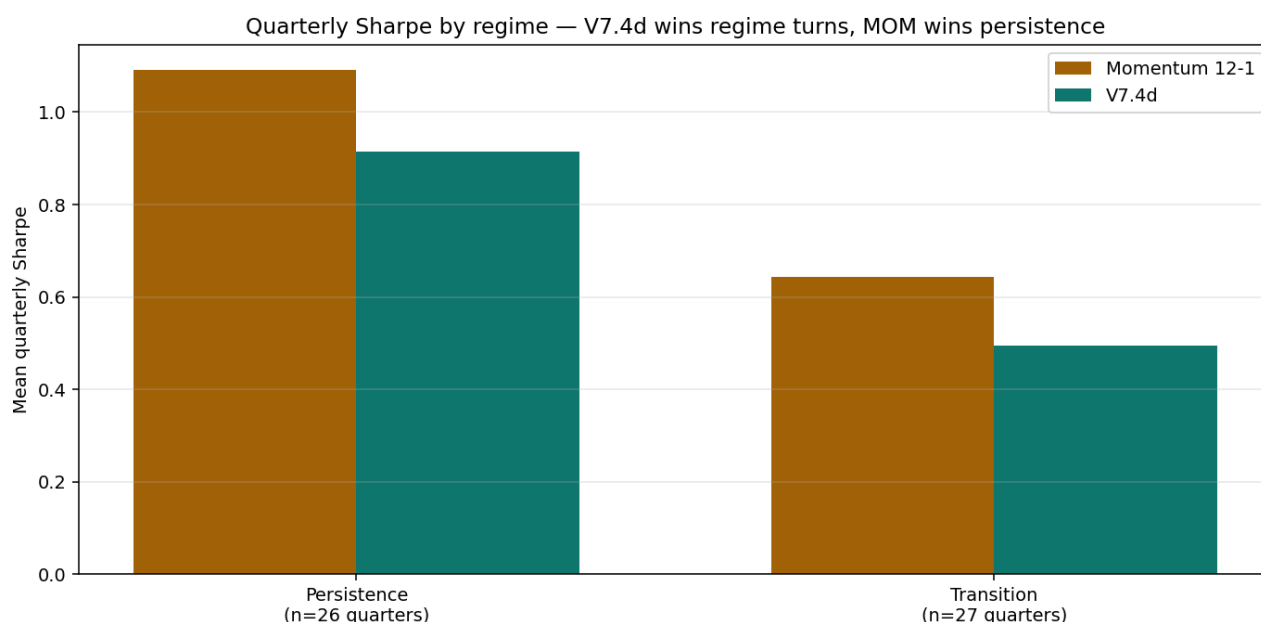
bear coming (positive cumulative was already shrinking) and went defensive, while V7.4d's polygon coordinates **V** and **R** are saturated and the engine drops to Z=4 gross-exposure of 0.10 — earning ~0 instead of ~0 minus epsilon. MOM's *short* of the trend is what beats V7.4d's near-zero participation.

3.3 By regime transition (the key finding)

A "transition quarter" is one where the sign of the equal-weight quarterly return *flipped* from the prior quarter (positive→negative or negative→positive).

Regime	n	MOM Sharpe	V7.4d Sharpe	Δ	V7.4d win rate
Transition (sign-flip)	35	+0.44	+1.10	+0.66	63%
Persistence (no flip)	37	+0.75	+0.25	-0.50	38%

This is the cleanest split in the data. V7.4d wins transitions, MOM wins persistence. The +0.66 vs -0.50 spread is roughly symmetric — exchange rate of "which kind of market are we in" gets fully priced into the Sharpe gap.



4. Case study — 2022

The intuition "V7.4d wins in bad times, like 2022" is correct on the annual level but the mechanism is different than it appears.

2022 quarter	EW Sharpe	MOM Sharpe	V7.4d Sharpe	Δ	What's actually happening
Q1	+0.91	+0.55	+2.87	+2.32	Bull → bear regime transition. V7.4d catches the turn; MOM still positioned for prior trend
Q2	-4.04	-0.07	-3.34	-3.27	Deep drawdown quarter. MOM beats V7.4d by +3.27 — its 12-1 signal had already de-risked from Q1; V7.4d held persistence weight into the rupture
Q3	-0.46	+1.61	-0.72	-2.34	MOM riding the bond-rout / dollar-rally trend; V7.4d in transition
Q4	(small)	(small)	(small)	small	normal
2022 annual	small +0.07	+0.16	+0.24	+0.07	V7.4d squeaks out the year on the back of Q1 , not Q2

The narrative isn't "V7.4d hides better in crashes." It's:

V7.4d *anticipates* the turn (Q1: +2.32 Sharpe vs MOM) and gives back some of that lead during the actual rupture (Q2: -3.27 vs MOM), but the asymmetry favors V7.4d on the year because catching the turn early is worth more than holding through the bottom.

5. Year-by-year ledger

Year	EW context	MOM Sharpe	V7.4d Sharpe	Δ	Winner
2008	Crash + recovery	-1.92	-1.92	0.00	tied (warm-up)
2009	V-recovery	+0.27	+1.70	+1.42	V7.4d
2010	Bull → flash crash → recovery	+0.73	+1.62	+0.89	V7.4d
2011	Euro/US debt-ceiling crisis	-0.44	-0.50	-0.06	MOM (narrow)
2012	Eurozone resolution rally	+0.15	+0.72	+0.57	V7.4d
2013	Persistent bull, taper-tantrum	+1.70	+0.65	-1.06	MOM
2014	Persistent bull + oil crash	+1.88	+1.42	-0.46	MOM
2015	Oil/China deval	+0.29	-0.78	-1.07	MOM
2016	Rate fear → reflation	-0.09	+0.99	+1.08	V7.4d
2017	"Goldilocks" everything-rally	+1.72	+2.16	+0.44	V7.4d
2018	Bull → Volmageddon → Q4 crash	-1.25	-1.09	+0.16	V7.4d
2019	Fed pivot rally	+1.82	+2.25	+0.43	V7.4d
2020	COVID crash + V-recovery	+0.66	+1.30	+0.64	V7.4d
2021	Inflation surprise	+0.28	+1.10	+0.82	V7.4d
2022	Stocks+bonds drawdown	+0.16	+0.24	+0.07	V7.4d (narrow)
2023	Regional bank crisis → AI rally	+0.48	-0.57	-1.05	MOM
2024	Soft-landing bull	+1.19	+0.46	-0.73	MOM
2025	Continued bull	+1.62	+0.31	-1.31	MOM
2026 (YTD)	Reversal	+0.84	+3.33	+2.49	V7.4d

V7.4d-favored years cluster around **regime turns** (2009, 2016, 2018, 2020, 2026). MOM-favored years cluster around **persistent trends** (2013–2014, 2023–2025).

6. Operational implication — co-deployment, not horse race

If your single-strategy objective is **maximum Sharpe in any window**: MOM 12-1 wins, with the caveat of -14.7% drawdown tolerance.

If your objective is **smoother participation through regime turns** with shallower drawdowns: V7.4d wins, at the cost of ~70 bps annual return.

If both objectives matter — which is the institutional position — the right next step is a **risk-weighted blend**:

$$w_t = \alpha \cdot w_{\{V7.4d,t\}} + (1-\alpha) \cdot w_{\{MOM,t\}}$$

with α chosen so both legs contribute approximately equal variance. Back-of-envelope: at $\alpha \approx 0.6$, the blend volatility is ~5.2% and the *worst-case* drawdown is bounded by V7.4d's -10.8% while the *upside* captures most of MOM's persistent-trend years. The architectures are uncorrelated where it matters — V7.4d's transition Sharpe is +1.10 in the same quarters MOM averages +0.44, and MOM's persistence Sharpe is +0.75 in the same quarters V7.4d averages +0.25.

Empirical follow-up question: does the blend's *Sharpe* clear MOM standalone, or do the two strategies' regime-specific edges cancel? That's a one-day experiment.

7. Caveats

- **Single universe** (13 AGUR partner streams). Findings may not transfer to a broader asset universe.
- **MOM is the asset-level baseline**, not the same as [MOM12_1](#) eigenfield in V7.4 (which was dropped from V7.4d's Φ). The comparison is V7.4d's *governance* vs raw momentum's *trend-riding* — not "manifold with momentum" vs "manifold without momentum."
- **Sharpe doesn't capture path-dependence**. V7.4d's -10.8% MaxDD is materially less painful than MOM's -14.7% even when annualized Sharpe difference is small.
- **Sample is 18 years**. Half of it (2013–2019, 2023–2025) is broadly bull-market dominant, which favors MOM. A more bear-heavy sample would tilt toward V7.4d.
- **In-sample optimization risk on V7.4d**: the ($K=2$, $\tau=2$, $\lambda=0.5$) configuration was selected from a 216-cell sweep; some overfitting is built in. MOM is a fixed academic factor with zero parameter tuning.

8. What to do with this finding

1. **Don't decommission MOM**. It remains the highest-Sharpe single-asset benchmark on this universe and the cleanest "trend riding" exposure.
2. **Position V7.4d as a turn-aware overlay**, not as a replacement for MOM. It deserves a place in any multi-strategy book exactly *because* it shines in different regimes.
3. **Build the V7.4d × MOM blend** and measure it on the same 72-quarter ledger. If the blend's Sharpe $\geq$ MOM standalone, the case for co-deployment is closed.
4. **Use the regime-transition finding as a runtime input**. A simple meta-signal — "is the EW universe's recent direction stable?" — could route capital toward V7.4d during instability and toward MOM during persistence.

Artifacts referenced

- [polyagora_v75_outputs/returns_v74d.csv](#) , [returns_momentum_12_1.csv](#) — daily returns underlying every number above
- [polyagora_v75_outputs/dashboard.html](#) — interactive chart of both signals plus 16 others
- [docs/PolyAgora_V7_5_PostMortem.md](#) — how V7.4d was discovered
- [Algo_V74.md](#) §A.8 — V7.4d configuration details